Metric topology

Prototype: Euclidean metric in $\mathbb{R}^n$

$\mathbb{R}^2$: $d(x_1, y_1, x_2, y_2) = \dots$

$\mathbb{R}^3$: $d(x_1, y_1, z_1, x_2, y_2, z_2) = \dots$

In $\mathbb{R}^n$, write $\bar{x} = (x^1, x^2, \dots, x^n)$ etc.

then $d(\bar{x}, \bar{y}) = \left((x^1-y^1)^2 + \dots + (x^n-y^n)^2\right)^{1/2}$

Open balls: $B_r(\bar{x}) = \{\bar{y} \in \mathbb{R}^n / d(\bar{x}, \bar{y}) < r\}$

Ex: In $\mathbb{R}^2$, $B_r(\bar{x})$ is ...

In $\mathbb{R}^3$, $B_r(\bar{x})$ is ....

Say $O \subset \mathbb{R}^n$ is open if for all $\bar{x} \in O$, there exist $r_{\bar{x}} > 0$ s/t $B_{r_{\bar{x}}}(\bar{x}) \subset O$.

So O is the union of all such $B_{r_{\bar{x}}}(\bar{x})$.

Aside: Need some properties of a metric:

(i) $d(\bar{x}, \bar{y}) \geq 0$, and $= 0$ if and only if $\bar{x} = \bar{y}$.

(ii) $d(\bar{x}, \bar{y}) = d(\bar{y}, \bar{x})$

(iii) $d(\bar{x}, \bar{z}) \leq d(\bar{x}, \bar{y}) + d(\bar{y}, \bar{z})$

where $\bar{x}, \bar{y}, \bar{z} \in \mathbb{R}^n$ are any.

What would you call (i), (ii), (iii)?

(Cauchy-Schwartz: $0 \leq |\,|\vec{u}|\vec{v} - |\vec{v}|\vec{u}\,|^2 = \dots$ expand using dot products)

How would you prove (i), (ii), (iii) for the Euclidean metric?

Note that (iii) reduces to the triangle inequality in $\mathbb{R}^2$!

Q: Is $B_r(\bar{x})$ open? Is $z \geq 0$ in $\mathbb{R}^3$ open?

In general, given a space X, a metric on X is a function $d: X \times X \to \mathbb{R}$ satisfying (i), (ii), (iii).

Use d to define a topology on X.

Ex: $X = \{x_1, x_2, \dots\}$
$d(x_i, x_j) = 1$ if $i \neq j$,
$d(x_i, x_i) = 0$.

Topology?

Ex: $X = \mathbb{R}^+$, $d(x,y) = |\ln(y/x)|$.

Topology? How does it compare with the Euclidean topology?

Ex: British railway metric
$d(x,y) = |x| + |y|$, $x \neq y$,
$d(x,x) = 0$.

$B_r(0)$?
$B_r(1)$?

Check that this defines a metric. Open balls?

Ex: $X = C[0,1]$ (continuous fncs on $[0,1]$). Let

$$d(f,g) = \max_{0 \le x \le 1} |f(x) - g(x)|$$

Check this is a metric! Open balls?

Ex: $X = C[0,1]$. Define

$$d(f,g) = \int_0^1 |f(x) - g(x)|\, dx$$

Check this is a metric! Open balls?

Questions:

1. Let O_α, $\alpha \in A$, be open in $\mathbb{R}^n$. Is $\bigcup_{\alpha \in A} O_\alpha$ open?

2. Is $\bigcap_{\alpha \in A} O_\alpha$ open?

Consider $O_n = (0, 1 + \frac{1}{n})$. Then $\bigcap O_n = (0, 1]$

But $O_1 \cap O_2 \cap \dots \cap O_k$ is open for any finite intersection. To wit, pick $\bar{x} \in O_1 \cap \dots \cap O_k$, then $B_{r_i}(\bar{x}) \subset O_i$, $r_i > 0$. Let $r = \min\{r_1, r_2, \dots, r_k\}$. Then $B_r(\bar{x}) \subset O_1 \cap \dots \cap O_k$.

$B_r(\bar{x}) \subset B_{r_i}(\bar{x}) \subset O_i$

Let X be a set. A topology on X is a subset $\tau \subset \mathcal{P}(X)$ (the power set of X) satisfying

(i) $\phi, X \in \tau$

(ii) If $\mathcal{O}_\alpha \in \tau$, $\alpha \in \mathcal{A}$, then $\bigcup_{\alpha \in \mathcal{A}} \mathcal{O}_\alpha \in \tau$.

(iii) If $\mathcal{O}_1, \mathcal{O}_2, \ldots, \mathcal{O}_n \in \tau$, then $\mathcal{O}_1 \cap \mathcal{O}_2 \cap \cdots \cap \mathcal{O}_n \in \tau$.

Ex: Indiscrete topology $\tau = \{\phi, X\}$.

Ex: Discrete topology $\tau = \mathcal{P}(X)$.

Ex: Sierpinski space: $X = \{0, 1\}$,
$\tau = \{\phi, \{1\}, \{0, 1\}\}$

Ex: $X = \{1, 2, 3\}$, $\tau = \{\phi, \{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

Ex: Any metric topology.

Call $N \subset X$ a nbhd of $x \in X$ if there is an open O s/t $x \in O \subset N$. Usually only consider open nbhds.

Write τ for a topology on X. So τ consist of a collection of subsets of X that are called open.

Let $Y \subset X$ be a subset of X, where X possesses topology τ. Define induced topology τ_Y on Y by

$$\tau_Y = \{ U \subset Y,\ U = Y \cap O,\ O \in \tau \}.$$

Also call τ_Y the subspace topology on Y. Check that τ_Y is a topology!

Ex: Let $\tau = \{ O \subset X,\ O \text{ any subset} \}$.
Discrete topology

Ex: Let $\tau = \{ \emptyset, X \}$. Indiscrete topology.

Call $C \subset X$ closed if $\neg C$ is open.
Then any intersection and finite unions of closed sets are closed. (Recall: De Morgan

$$\neg \bigcup_{\alpha \in A} Y_\alpha = \bigcap_{\alpha \in A} \neg Y_\alpha,$$
$$\neg \bigcap_{\alpha \in A} Y_\alpha = \bigcup_{\alpha \in A} \neg Y_\alpha.)$$

Ex: $\overline{B_r(\bar{x})} = \{\bar{y} \in \mathbb{R}^n \mid d(\bar{x},\bar{y}) \le r\}$ is closed.

Ex: What are the closed sets in discrete topology?

Ex: In $\mathbb{R}^2$, let $Y = \{(x,y) \mid x \ge 0, y > 0\}$
Open or closed in the Euclidean topology?

Let (X,τ) be given, and let $A \subset X$. Say $p \in X$ is an accumulation (or limit) point of A if every nbhd of p intersects $A \setminus \{p\}$.

Ex: $A = \{1, 1/2, 1/3, \dots\} \subset \mathbb{R}$. Acc. pnts?

Ex: $A \subset \mathbb{R}^n$, $(x^1, \dots, x^n) \in A$ if $x^i \in \mathbb{Q}$. Acc. pnts?

Ex: $(x^1, \dots, x^n) \in \mathbb{R}^n$, x^i is an integer. Acc. pnts?

Thm: A set $Y \subset X$ is closed iff it contains all its limit points.

Pf: Suppose Y is closed. Then $\neg Y$ is open, so any $p \in \neg Y$ has a nbhd U s/t $U \cap Y = \emptyset$. Thus p can not be a limit point, so Y contains all its limit points.

Next assume that Y contains all its limit points. Pick $p \in \neg Y$, then p is not a limit point of Y.... so $\neg Y$ must be open.

Def'n: The union of A and all its limit points is called the closure $\bar{A}$ of A.

Thm: $\bar{A}$ is the smallest closed set containing A.

Pf. Start with $\bar{A}$ is closed:
Pick $p \in \neg\bar{A}$. Then p is not a limit point of A so there is an open $U \subset X$ s/t $p \in U$ and $U \cap A = \phi$. Now U is also a nbhd of any of its points so $U \cap \bar{A} = \phi$. It follows that $\neg\bar{A}$ is open so $\bar{A}$ is closed.

Next let $A \subset B$, B closed. Then any limit point of A is also a limit point of B, and so must be contained in B since B is closed. Hence $\bar{A} \subset B$
Thus $\bar{A} = \cap B$, $A \subset B \subset X$, B closed.

Suppose $A \subset X$ is such that $\bar{A} = X$
Then call A dense in X.

Ex: $\mathbb{Q} \subset \mathbb{R}$
Ex: $\{(x,y) \in \mathbb{R}^2 \mid x,y \in \mathbb{Q}\}$

The interior A° of A is the union of all open sets contained in A.

Ex: $\mathbb{Q} \subset \mathbb{R}$ Interior?
Ex: $\mathbb{Q} \times \mathbb{Q} \subset \mathbb{R}^2$ Interior?
Ex: $D = \{(x,y) \mid x^2+y^2 \leq 1\} \subset \mathbb{R}^2$ Interior?

The frontier (boundary) ∂A of A is $\overline{A} \cap \overline{X \setminus A}$. So any nbhd $p \in \partial A$ must intersect both A and $X \setminus A$.

Ex: $\overline{B_1(0)} \subset \mathbb{R}^3$
Ex: $\mathbb{Q} \subset \mathbb{R}$

Basis for τ on X: A set $\beta \subset \tau$ s/t any $O \in \tau$ is a union of elements in β.

Ex: $\mathbb{R}$, $\beta = \{$all open intervals$\}$
Ex: $\mathbb{R}$, $\beta = \{$all open intervals with rational endpoints$\}$
This is countable. (Why is $\mathbb{Q}$ countable?)

Thm: Let $\beta \subset \mathcal{P}(X)$ be a collection of subsets of X. Suppose that β satisfies the conditions:
(a) For every $p \in X$ there is $B \in \beta$ s/t $p \in B$;
(b) Let $p \in B_1 \cap B_2$, where $B_1, B_2 \in \beta$. Then there is $B_3 \in \beta$ s/t $p \in B_3 \subset B_1 \cap B_2$.
Then β (together with ϕ) is a basis for a topology.